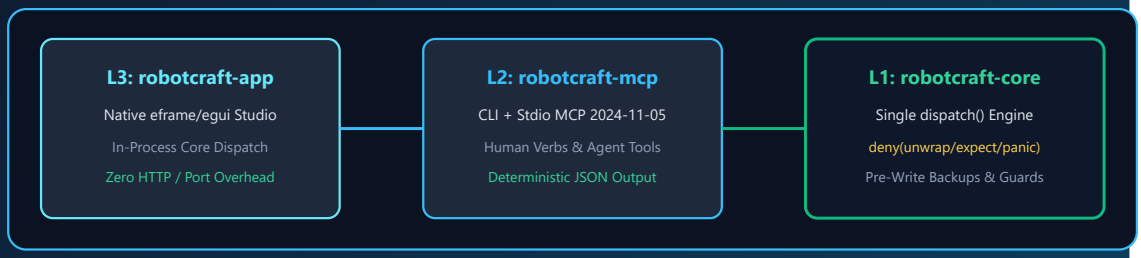
ARTCRAFT FLEET TECHNICAL BROCHURE · SERIES #15

RobotCraft v0.1.0

RobotCraft is the embodied intelligence studio and universal robotics operating workstation for the sandraschi fleet. It consolidates hardware telemetry, forward/inverse kinematics solvers, URDF models, Gazebo simulation bridges, and physical actuator tele-op.



v0.1.0	4 Crates	2166	6-Gate
RELEASE VERSION	WORKSPACE TIERS	PURE-RUST LOC	ARTCRAFT VERIFIED

Repository: `D:\Dev\repos\robotcraft` · **Standard:** `CRAFT_BROCHURE_STANDARD.md` · **Compiled via:** `BookCraft v0.5.0`

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Architecture

- **Consolidates Fleet Repos**: `robotics-mcp`, `ros-mcp`, `gazebo-mcp`, `unitree-mcp`, `yahboom-mcp`, `limx`, `nori` (`norirobotics-mcp`: Nori A3 home robot, LeRobot recording, VR spawn), `vbot`, `rust-saschirob-mcp` (personal robotics cookbook).
- **Universal 3-Tier ArtCraft Standard**:
- `crates/robotcraft-hal`: Hardware Abstraction Layer (Unitree UDP, Yahboom serial, ROS 2 DDS, Gazebo).
- `crates/robotcraft-core`: Kinematics solvers, URDF link trees, and Command Bus (`Command::dispatch`).
- `crates/robotcraft-mcp`: Stdio JSON-RPC server (MCP 2024-11-05) + CLI (`robotcraft-cli`).
- `crates/robotcraft-app`: Immediate-mode Native Studio GUI (`robotcraft`).
- **Standard Control Port**: `14116`.

Status — does today / next week / non-goals

Does today: 4-crate Rust workspace (core/HAL split) + web viewer; URDF/FK/IK verbs declared; 3D viewer rendering landed; 2 test fns.

Next week: PRD (HAL traits, sim choice, tele-op confirm gates, voice-bus paragraph); FK/IK fixture tests (sim-only CI); fk-sim recipe.

Non-goals: no physical actuation without confirm gates; hardware matrix stays gated, never mock-green.

Lineage — the Storytold craft-app pattern

Storytold ArtCraft (storytold/artcraft — "the IDE for artists") was the starting point. The 16 craft apps in the Storytold stable each ship one binary that is both the human CLI and the agent MCP server (stdio), headless-first, with `run.json` + report artifacts. That modus operandi and harness is what this repo implements; the code here is independent, the pattern is shared.